

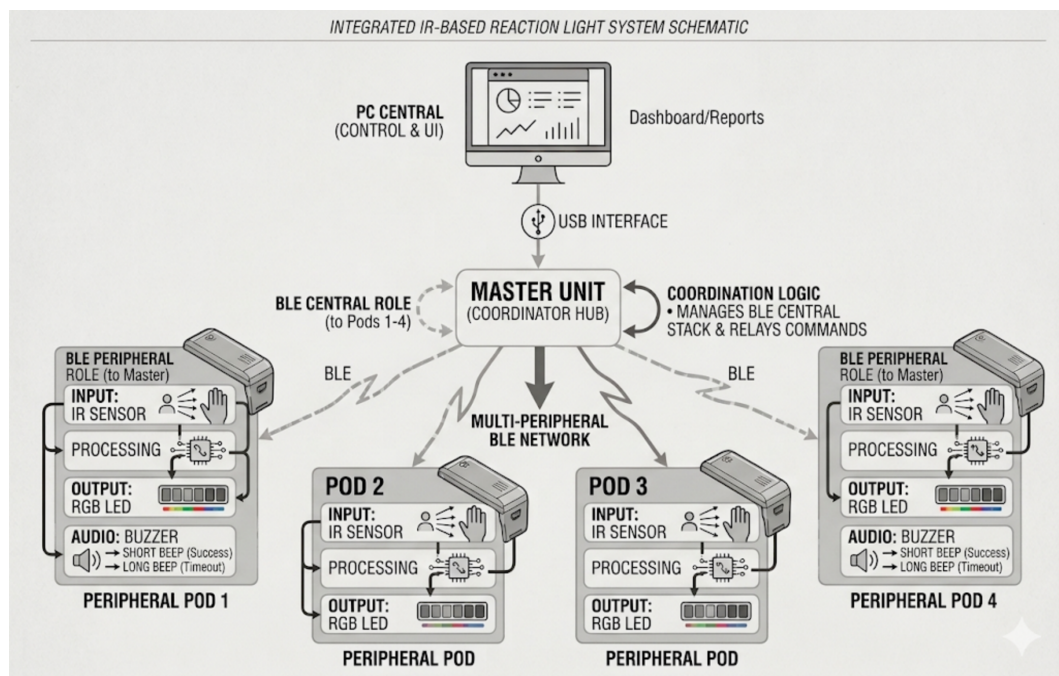
Cognitive Strength & Coordination System

Objective

The objective of this project is to develop a high-precision, IR-based reactive network designed to enhance cognitive processing speed and neuromuscular coordination.

By utilizing a synchronized system of four peripheral pods and a central PC-linked master hub, the system creates a training environment that challenges executive functions such as selective attention and decision-making through real-time visual and auditory stimuli. The system measures and improves the user's ability to process complex cues and execute rapid motor responses without physical contact.

Architecture Diagram



Key Features

- 1 Non-Contact Infrared (IR) Proximity Triggering
- 2 High-Precision Reaction Timing
- 3 Cognitive Load RGB Logic
- 4 Dual-Tone Auditory Feedback
- 5 Asynchronous BLE Peripheral Management
- 6 USB-Serial Master Interface
- 7 Centralized Desktop Control
- 8 Mechanical Wear Reduction